Project

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Background

Problem Formulation

The NBA Draft is an annual event where all 30 NBA teams select eligible players to join their rosters, serving as the primary pathway for new talent to enter the league. The draft consists of two rounds, with each team holding one pick per round, for a total of 60 selections. The order in which teams pick is not arbitrary: teams that performed worse in the previous season are rewarded with earlier picks, giving struggling franchises the opportunity to rebuild through high-potential young talent. Specifically, the 14 lowest-ranked teams enter a weighted lottery to determine the first 14 picks of the first round, with worse records receiving higher odds. The remaining picks in both rounds follow reverse standings order — the worst team picks first, the best team picks last. Prior to the draft, teams evaluate prospects through the NBA Draft Combine, private workouts, and scouting reports. Scouts assess a wide range of factors including physical measurements, athletic testing, skill level, and projected development trajectory. The result is an internal team ranking that ultimately determines which player a team selects when their pick arrives.

Pick position carries substantial consequences for a player's career. First-round selections receive fully guaranteed multi-year contracts, whereas second-round picks are not guaranteed and must compete for roster spots with little job security. This distinction makes the first-round versus second-round boundary a meaningful and practically important threshold to study.

Given this context, understanding what physical attributes are associated with first-round selection is of both scientific and practical interest. Height, in particular, has long been regarded as a fundamental advantage in basketball — taller players are better positioned to score near the basket, contest shots, and control the boards. It is therefore natural to ask whether teams systematically favour taller players when making first-round selections. Our scientific question is:

> **Does player height have a positive effect on the probability of being selected in the first round of the NBA Draft?**

Mathematically, we model this via Bayesian logistic regression, where the binary outcome (first round = 1, second round = 0) is regressed on player height. The key quantity of interest is the **posterior probability that the slope on height is positive**:

$$P(\beta_{\text{height}} > 0 \mid \mathbf{Y})$$

Team Contributions

Jenni(40%) is primarily responsible with models, such as model development, stan code, model set up, model diagnostics, data cleaning, and model selection. Yasmine(30%) is primarily responsible with the dataset and

goddness of fit, such as explorative data analysis, calibration, plot interpretation, background introduction, problem formulation, data cleaning and table interpretation. Max(30%) is responsible with finding the actual data, formulating problem, and worked on literature review, discussion, limitation, reference, and result comparison. We all contributed significant amount of time and effort in the project.

Public repository link: https://github.com/ymyms/stat-405-project

Literature Review

A growing body of literature examines how NBA teams evaluate talent and make draft decisions, focusing on player performance and observable characteristics. A closely related study is Greer et al. (2019), which uses a logistic regression framework to analyze the determinants of NBA draft selection. The authors find that scoring ability is the most important predictor of whether a player is selected, while other factors such as team success and player characteristics also play a role. Complementing this, Cui et al. (2019) analyze NBA Draft Combine data and show that anthropometric characteristics, particularly height and wingspan, are key factors distinguishing drafted from undrafted players. Their findings highlight the importance of physical attributes in talent identification and suggest that taller players have a systematic advantage in draft outcomes.

Despite these contributions, existing studies primarily focus on whether a player is drafted at all, rather than distinguishing between first-round and second-round selection, which represents a meaningful economic threshold due to differences in contract guarantees. Furthermore, most prior research relies on frequentist methods and emphasizes statistical significance rather than probabilistic interpretation. As a result, these approaches do not directly quantify the probability that a specific attribute, such as height, has a positive effect on draft outcomes.

This paper contributes to the literature by focusing on the probability of being selected in the first round and adopting a Bayesian logistic regression framework. This approach allows for direct estimation of the posterior probability that height has a positive effect on draft selection, providing a more intuitive and decision-relevant interpretation. Using a publicly available NBA player dataset from Kaggle, this study offers a complementary perspective on how physical attributes influence draft outcomes.

Variable Introduction

The primary variables which might associate with draft round are

- Age
- Player Height (cm)
- Player Weight (kg)
- Draft year
- The number at which the player was picked in his draft round
- Games played throughout the season
- Average number of points scored
- Average number of assists distributed
- Percentage of available offensive rebounds the player grabbed while he was on the floor

Explorative Data Analysis

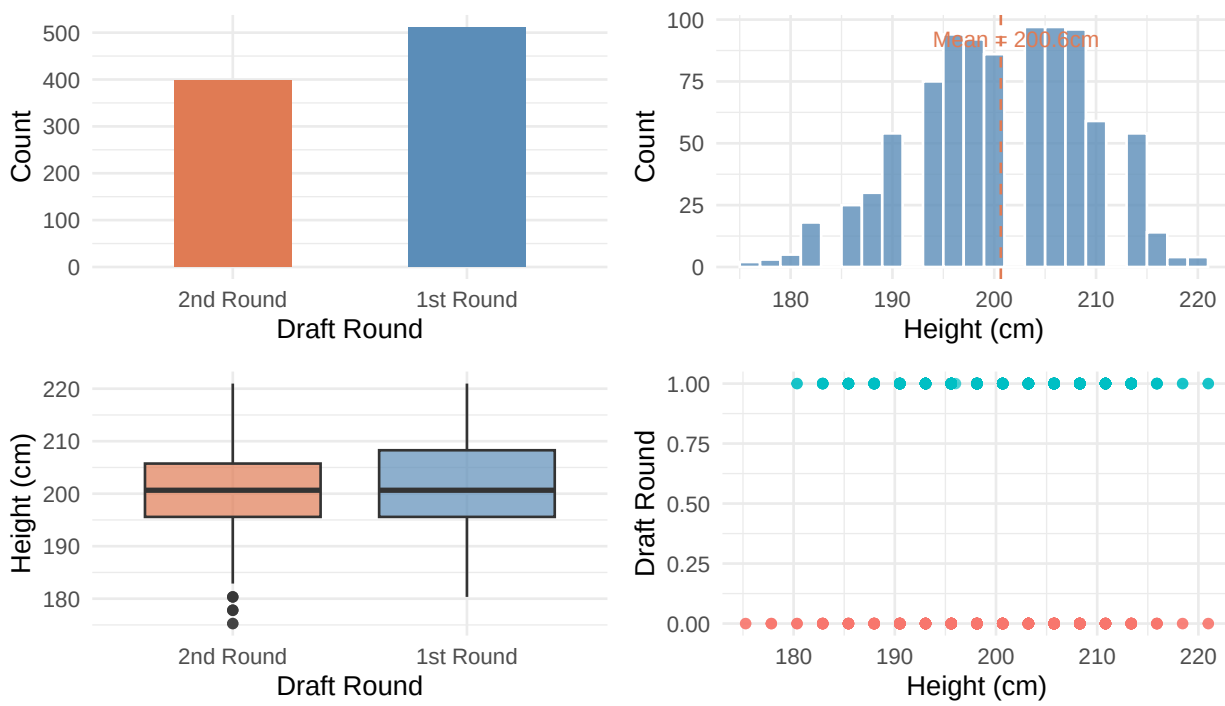
The goal of this section is to examine the distribution of player height and its relationship with draft round.

Data Cleaning

The raw dataset contains 12,844 player-season observations across 22 variables, spanning the 1996–97 to 2022–23 NBA seasons. Each row represents one player in one season, meaning the same player may appear multiple times across different years.

Four cleaning steps were applied to prepare the data for modelling. First, we restricted the data to players drafted in 2006 or later, retaining only the past two decades of draft classes. This ensures that our findings reflect the preferences and evaluation criteria of the modern NBA. Second, players recorded as undrafted were excluded, since our research question concerns the distinction between first- and second-round picks. Specifically, undrafted players represent a qualitatively different pathway into the league. Third, to avoid the statistical issue of repeated observations per player (the same player's height and draft round appearing across multiple seasons), we retained only each player's earliest season. This gives one observation per unique player, ensuring the independence assumption of logistic regression. Fourth, the `draft_round` variable was recoded as a binary indicator: 1 for first-round selections and 0 for second-round selections.

Distribution of Draft Round and Height




```
## # A tibble: 2 x 5
##   `Draft Round`     N  Mean    SD Median
##   <chr>         <int> <dbl> <dbl>  <dbl>
## 1 2nd Round       397  200.  8.63   201.
## 2 1st Round       512  201.  8.16   201.
```

From the dot plot, we can see that when player's height is below 180cm, it is very unlikely for the player to be drafted in the first round. After 180cm, the nuisance becomes less obvious. From the boxplot, we can see that the first-round picks are on average taller than second-round picks, though the difference is modest and the distributions overlap considerably. The EDA is consistent with the prior findings by Cui et al, that anthropometric characteristics play a key role in drafts.

Regression Model Selection

In order to select the optimal logistic regression model, we performed a forward selection method using adjusted R^2. The logistic model intentionally excluded player name, team, college, and other variables which have too many levels that may introduce too many linear dependencies.

The model with the maximized adjusted R^2 contains seven covariates, which are `age`, `player_height`, `player_weight`, `draft_number`, `pts`, `ast`, and `oreb_pct`.

Bayesian Model

We would like to propose two models- a regular model and a hierarchical model. We would use two methods to compute the posterior: Hamiltonian Monte Carlo and Variational Inference. HMC using stan is very fast and efficient, and VI works well for our large dataset with 909 observaions. We would compare and contrast between the methods for each model.

Regular Bayesian Model

The model has distribution as following.

$$\text{intercept} \sim N(0, 1)$$

$$\beta_i \sim N(0, \sigma_i)$$

$$\theta = \text{intercept} + \text{height} \cdot \beta_{\text{height}} + \text{weight} \cdot \beta_{\text{weight}} + \text{pts} \cdot \beta_{\text{pts}} + \text{ast} \cdot \beta_{\text{ast}} + \text{draft number} \cdot \beta_{\text{draft number}} + \text{age} \cdot \beta_{\text{age}} + \text{pct oreb} \cdot \beta_{\text{pct oreb}}$$

$$Y \sim \text{Bernoulli}(\textbf{logistic}(\theta))$$

For this model, σ_i is assigned different value, for sake of sensitivity analysis. We used three different choices of priors of standard deviation for comparison.

- For $\sigma_{\textbf{height}} = \sigma_{\textbf{height}} = 3, \sigma_{\textbf{pts}} = \sigma_{\textbf{ast}} = 2, \sigma_{\textbf{draft number}} = \sigma_{\textbf{age}} = \sigma_{\textbf{pct oreb}} = 1$, we are assuming that the anthropometric characters are the most variant.
- For all $\sigma_i = 1$, we are assuming that the all coefficients have the same standrd normal distribution.
- For $\sigma_{\textbf{height}} = \sigma_{\textbf{height}} = 1, \sigma_{\textbf{pts}} = \sigma_{\textbf{ast}} = 0.5, \sigma_{\textbf{draft number}} = \sigma_{\textbf{age}} = \sigma_{\textbf{pct oreb}} = 0.1$, we are assuming that the coefficients are not as variant as suggested in the first prior suggestion.

Our result is as follows, computed using HMC method:

	$E[\beta_{height}\|Y]$	$P[\beta_{height} > 0\|Y]$	95% CI for β_{height}
Prior Choice 1	0.149	1.000	[0.105, 0.192]
Prior Choice 2	0.142	1.000	[0.102, 0.182]
Prior Choice 3	0.099	1.000	[0.071, 0.129]

Our result for variational inference is:

	$E[\beta_{height}\|Y]$	$P[\beta_{height} > 0\|Y]$	95% CI for β_{height}
Prior Choice 1	0.253	0.993	[0.051, 0.456]
Prior Choice 2	0.251	0.993	[0.052, 0.450]
Prior Choice 3	0.131	1.000	[0.0618, 0.200]

From the two tables, we can conclude thay for VI, the posterior means for β_{height} are larger than that computed by HMC, but the posterior probabilities of $\beta_{height} > 0$ are usually smaller than HMC's. However, the credible interval for VI are wider, indicating more uncertainty compared to HMC.

For both methods, changing the prior, we can see that the output do vary. When standard deviation is assumed smaller, $E[\beta_{height}|Y]$ tend to be smaller. It seems like both methods are somewhat sensitive to the choice of prior. However, one thing they both agree on is that $P[\beta_{height} > 0|Y]$ usually round to 1, meaning that the slope is almost always positive. The 95% credible interval β_{height} does not contain 0, which supported with this conlcusion. Therefore, for our research interest, both methods are sufficient and efficient.

Hierarchical model

This hierarchical model differ from the regular ones, wher individuals might be in specific groups. For sake of this project, we use hierarchical model, assuming that coefficients for different parameters might have the same distribution.

The model has distribution as following.

$$\text{intercept} \sim N(0, 1)$$

$$\text{center} \sim N(\mu, \sigma)$$

$$\text{spread} \sim Exp(\lambda)$$

$$\beta_i \sim N(\text{center}, \text{spread})$$

$$\theta = \text{intercept} + \text{height} \cdot \beta_{\text{height}} + \text{weight} \cdot \beta_{\text{weight}} + \text{pts} \cdot \beta_{\text{pts}} + \text{ast} \cdot \beta_{\text{ast}} + \text{draft number} \cdot \beta_{\text{draft number}} + \text{age} \cdot \beta_{\text{age}} + \text{pct oreb} \cdot \beta_{\text{pct oreb}}$$

$$Y \sim \text{Bernoulli}(\textbf{logistic}(\theta))$$

For this model, μ, σ, λ are parameters we can change for sake of sensitivity analysis.

- For $\mu = 0, \sigma = 1, \lambda = 1$, we are assuming less variant center, and moderate spread.
- For $\mu = 0, \sigma = 10, \lambda = 10$, we are assuming variant center, and small and concentrated spread.
- For $\mu = 10, \sigma = 1, \lambda = 1/10$, we are assuming less variant but larger center, and laregr and variant spread.

Our result is as follows, computed using HMC method:

	$E[\beta_{height}\|Y]$	$P[\beta_{height} > 0\|Y]$	95% CI for β_{height}
Prior Choice 1	0.138	1.000	[0.098, 0.182]
Prior Choice 2	0.134	1.000	[0.094, 0.178]
Prior Choice 3	0.152	1.000	[0.108, 0.199]

Our result for variational inference is:

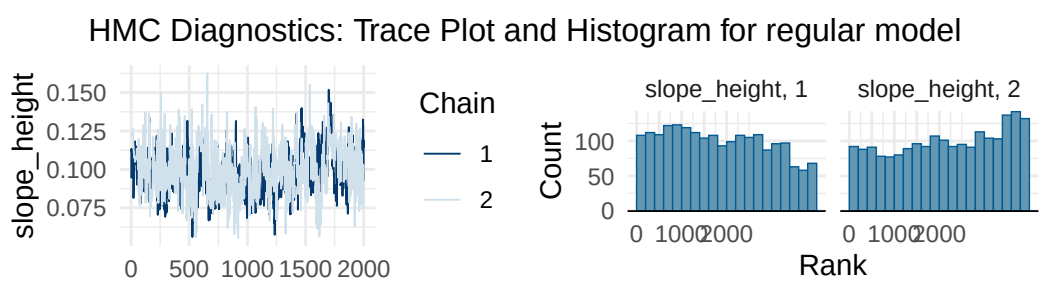
	$E[\beta_{height}\|Y]$	$P[\beta_{height} > 0\|Y]$	95% CI for β_{height}
Prior Choice 1	0.248	0.993	[0.061,0.444]
Prior Choice 2	0.244	0.992	[0.055, 0.436]
Prior Choice 3	0.256	0.993	[0.056,0.460]

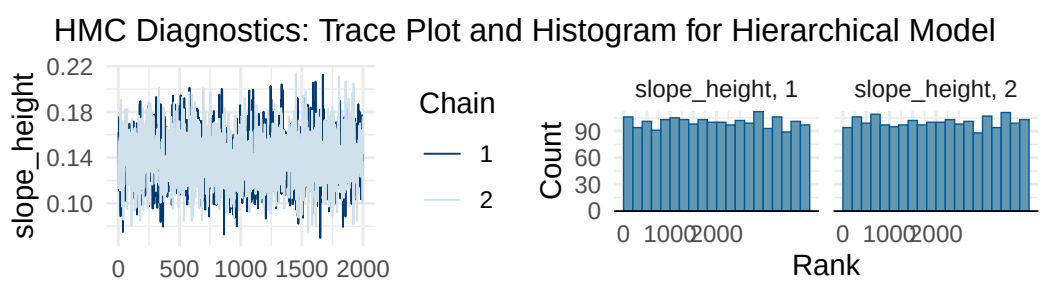
For this model too, VI has larger posterior means for β_{height} and smaller posterior probabilities of $\beta_{height} > 0$ than HMC's.

For both methods, changing the prior varies output, but only by a little. It seems like both methods are less sensitive to the choice of prior. However, one thing they both agree on is that $P[\beta_{height} > 0|Y]$ usually round to 1, meaning that the slope is almost always positive. The 95% credible interval for β_{height} does not contain 0, which supported with this conlcusion.

MCMC Diagnostics

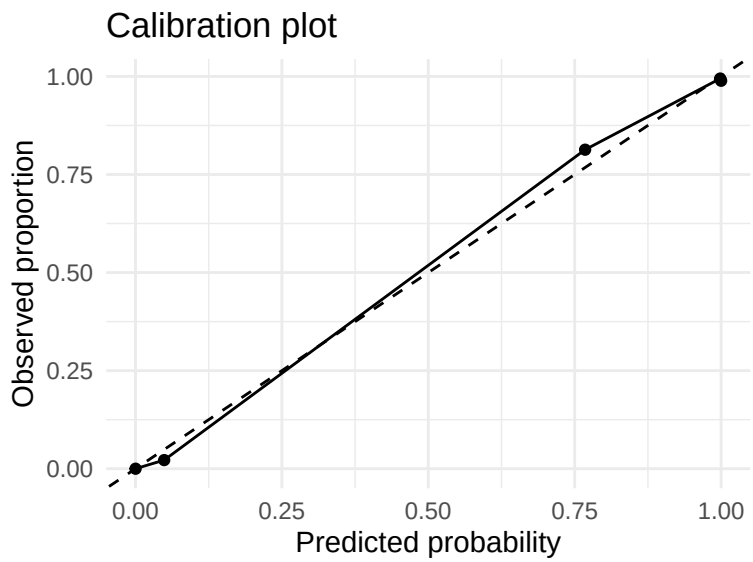
Mixing





We choose to use 2 chains so we could check whether the chain mixes well. Here, we are using chains for `slope_height` to check whether the chain mixes well. `height_slope_positive` is binary, which could be assessed, but would show less chain movements than continuous models. Ensuring that both variables are form the same model and method, if the plots for `slope_height` suggests fast mixing, then it means that the model for `height_slope_positive` mixes fast. From both trace plots, we can see that the two chains overlap a lot, showing a carterpillar shape; from the histogram, we can see two approximately uniform histograms. Both plots suggests that our model is fast mixing.

Goodness of Fit



For this calibration plot, we can see that the our actual coverage align closely with nominal coverage, which suggests that our model is well-specified, so calibration holds.

Validity of VI

CmdStan diagnose suggests that VI for both models have $\bar{R}$'s close to 1 and effective sample sizes satisfactory for all parameters. This suggests that VI provide stable posterior result. One thing to note is that this diagnose does not check the quality of VI itself. But evaluating the comparison between HMC and VI, we see that they agree, so VI is likely well-specified.

References

Cirtautas, J. (2023, October 13). NBA players. Kaggle. https://www.kaggle.com/datasets/justinas/nba-players-data/code

Cui, Y., Liu, F., Bao, D., Liu, H., Zhang, S., & Gómez, M.-Á. (2019). Key anthropometric and physical determinants for different playing positions during National Basketball Association Draft Combine Test. Frontiers in Psychology, 10. https://doi.org/10.3389/fpsyg.2019.02359

Greer, T., Price, J., & Berri, D. (2019). Jumping in the pool: What determines which players the NBA considers in the draft? International Journal of Sport Finance, 14(1), 44–54. https://doi.org/10.32731/ijsf/141.022019.04